

1) ICAR – Central Institute for Research on Cotton Technology (CIRCOT), Mumbai

📍 Location: Matunga, Mumbai

🔗 Focus Areas:

Cotton fiber quality testing and standards

Ginning and spinning technology

Cotton by-product processing (e.g., oil, linters)

Natural dyes, bio-scouring, nanotech applications

🔗 Key Innovations:

Pre-grooved roller technology: Boosted lint yield by 10%

Portable ginning machines: Lilliput Gin, CLOY, Hipro – ideal for small farms

MoUs & Patents: Nano-zinc storage solutions, antimicrobial bags for cotton storage

📖 Training & Research Outputs:

Manuals and training programs in English, Hindi, and Marathi

Postgraduate & PhD programs affiliated with Mumbai University

Extension videos and bulletins for farmers

🗣️ How it helps your project:

Use these documents in your RAG-based chatbot to answer queries like:

“Which roller improves cotton quality?”

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🏆 Achievements:

61+ rice varieties released for Konkan conditions

Yields increased from 1–1.2 t/ha (1970s) → 3–7.5 t/ha

Developed hybrids like Sahyadri-1/4 and “Karjat-10” (2024)

📖 Data for Chatbot:

Local advisories in Marathi and English

Production guides, fertilization schedules, pest management plans

🗨️ Sample multilingual queries:

“What is the yield of Karjat-7?”

How to Use This in Your Chatbot

Step-by-step integration:

Collect Documents – From CIRCOT and DBSKKV official websites or ICAR repositories

Extract Text – Use OCR (e.g., PyMuPDF, Tesseract)

Translate – Between English, Hindi, Marathi using APIs

Store in Vector DB – Use FAISS, Chroma, or Pinecone

Connect to LLM – Use LangChain/RAG pipeline to answer queries contextually

Test Multilingual Chatbot – Check outputs across languages and use cases

“कॉटनची गुणवत्ता सुधारण्यासाठी कोणती यंत्रे?”